

HPHT ENERGISED FOAM

PhD Proposal • Supervisor Discussion

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Research Problem & Proposed Workflow

Presentation Overview

1

Research Problem Architecture

What the research problem is, and how physics, prediction and engineering decisions connect.

2

Proposed Research Workflow

How the research will actually be carried out, from data to engineering assessment.

3

How the Workflow Answers the Questions

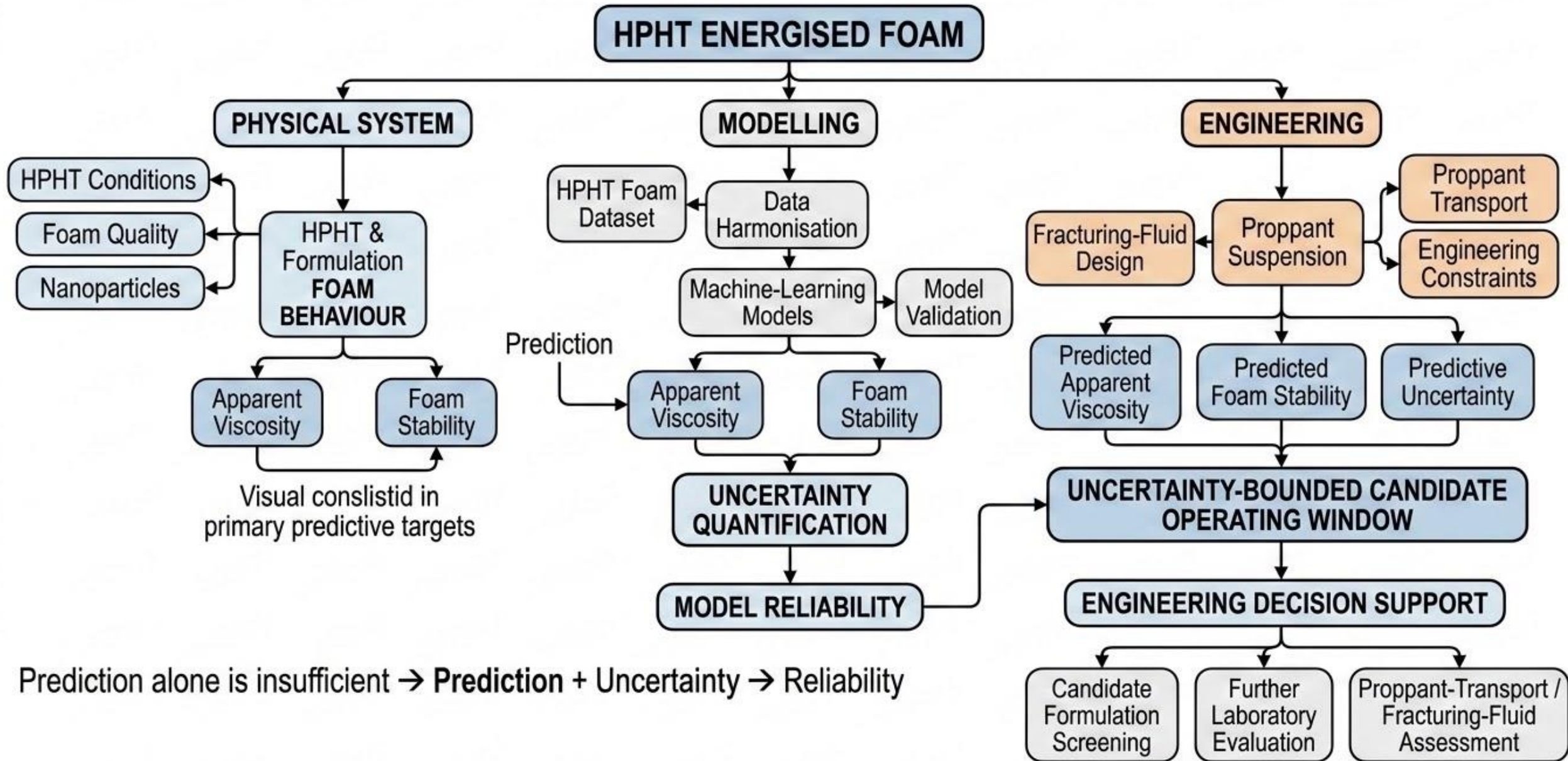
Mapping the two diagrams onto the supervisor's core concerns.

4

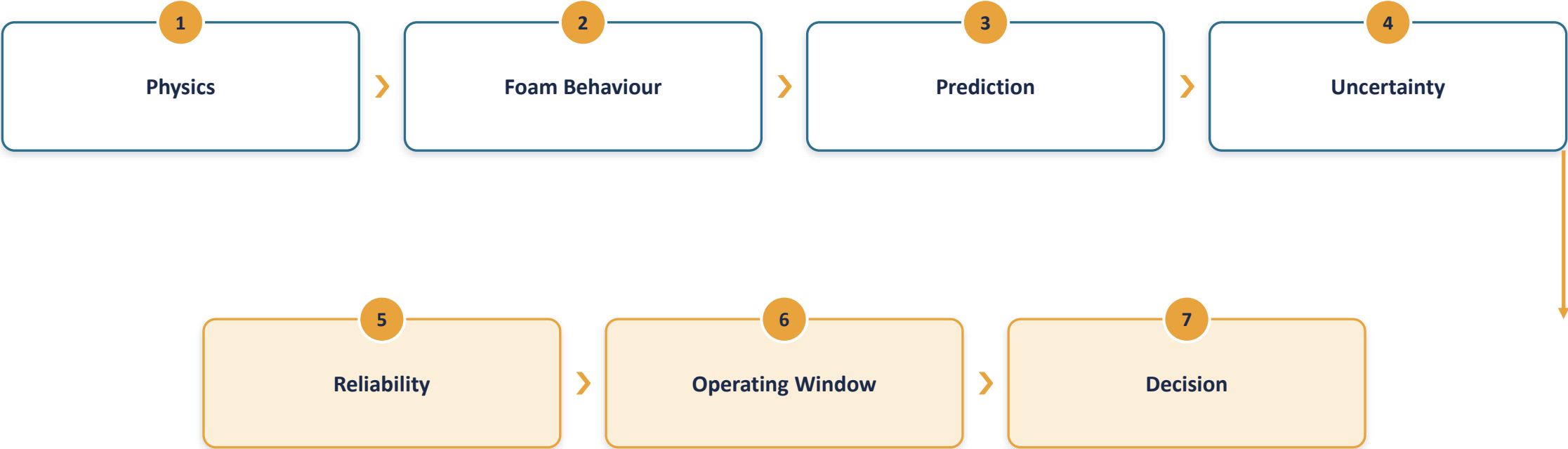
Discussion

Open questions, feedback and next steps.

HPHT Energised Foam — Research Problem Architecture



Research Problem Architecture



Physics → Foam Behaviour → Prediction → Uncertainty → Reliability → Operating Window → Decision

Purpose: Explains what the research problem is and how its main components connect — from HPHT foam physics to modelling, uncertainty, and engineering decisions.

Key Points & Supervisor Questions

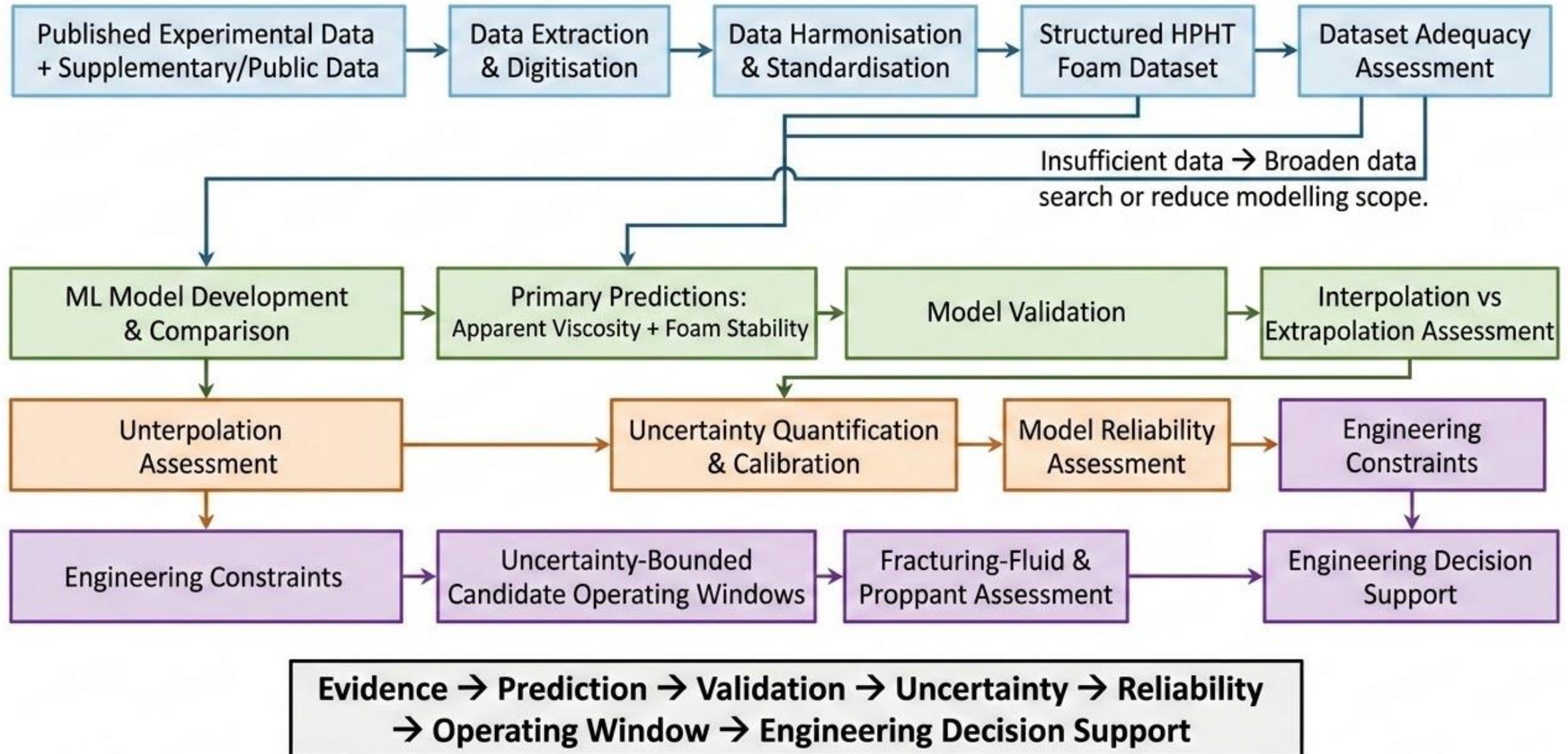
KEY POINTS

- ✧ The physical system studied is HPHT energised fracturing foam.
- ✧ HPHT conditions, foam quality and formulation control apparent viscosity and foam stability.
- ✧ Apparent viscosity and foam stability are the primary predictive targets.
- ✧ Machine learning is used to capture the complex nonlinear relationships between them.
- ✧ Uncertainty is quantified because prediction alone does not establish reliability.
- ✧ Reliable predictions are combined with engineering constraints to identify candidate operating windows.

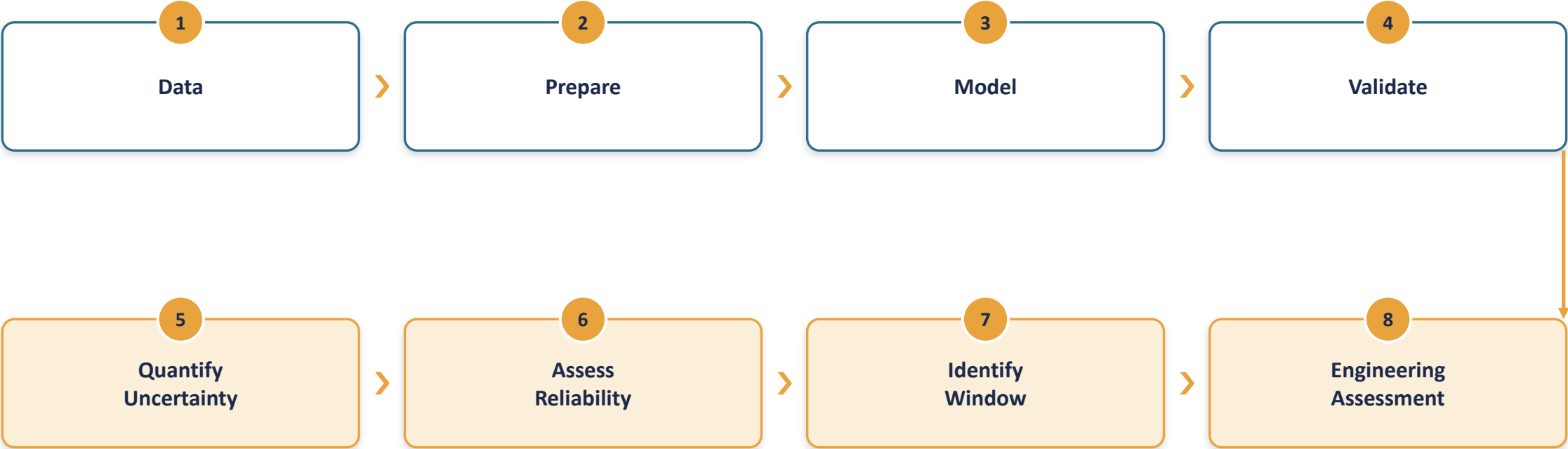
SUPERVISOR QUESTIONS THIS HELPS ANSWER

- ✧ Understanding of HPHT foam rheology and hydraulic fracturing-fluid design
- ✧ How uncertainty quantification would be implemented
- ✧ How operating windows link to engineering decision-making

HPHT Energised Foam — Proposed Research Workflow.



Proposed Research Workflow



Data → Prepare → Model → Validate → Quantify Uncertainty → Assess Reliability → Identify Window → Engineering Assessment

Purpose: Explains how I will actually carry out the research — from data acquisition and harmonisation through modelling, validation, uncertainty quantification and reliability assessment to operating-window identification.

Key Points & Supervisor Questions

KEY POINTS

- ✧ Begin with published experimental, supplementary and available research data.
- ✧ Extract, digitise, harmonise and standardise the data into a structured HPHT dataset.
- ✧ Assess whether the dataset is adequate before deciding the final modelling complexity.
- ✧ Develop ML models to predict apparent viscosity and foam stability.
- ✧ Validate the models, distinguishing interpolation from extrapolation, and quantify uncertainty.
- ✧ Combine reliable predictions with engineering constraints to identify candidate operating windows.

SUPERVISOR QUESTIONS THIS HELPS ANSWER

- ✧ The likely data sources for the proposed modelling work
- ✧ How you would manage limited and heterogeneous HPHT foam datasets
- ✧ How uncertainty quantification would be implemented in practice
- ✧ How operating windows would be linked to engineering decision-making
- ✧ Your technical preparation for carrying out the proposed work

How the Workflow Answers the Research Questions

Supervisor Question	Diagram 1	Diagram 2
Understanding of HPHT foam rheology & fracturing-fluid design	✓	
Likely data sources & managing heterogeneous datasets		✓
How uncertainty quantification would be implemented	✓	✓
How operating windows link to engineering decision-making	✓	✓
Technical preparation for carrying out the proposed work		✓

Both diagrams converge on the same reliability-driven pathway from prediction to engineering decision.

KEY TAKEAWAY

Prediction → Uncertainty → Reliability → Decision

1

Predicts foam apparent viscosity and stability under HPHT conditions.

2

Quantifies uncertainty to assess prediction reliability.

3

Translates reliable predictions into candidate operating windows — supporting engineering judgement, not replacing it.

Discussion

I welcome your feedback on the research scope, methodology, and proposed direction.